

PLANT DISEASE CLASSIFICATION USING CNN DEEP LEARNING MODEL

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PROJECT OBJECTIVE

>To classify plant leaf diseases using Deep Learning.

>To identify 15 different plant classes.

>To train a CNN model on plant leaf images.

>To predict the disease from a leaf image.

DATASET

- Dataset: PlantVillage
- Total Images: 20,638
- Total Classes: 15

- Training Images: 16,511
- Validation Images: 4,127

DATA PREPROCESSING

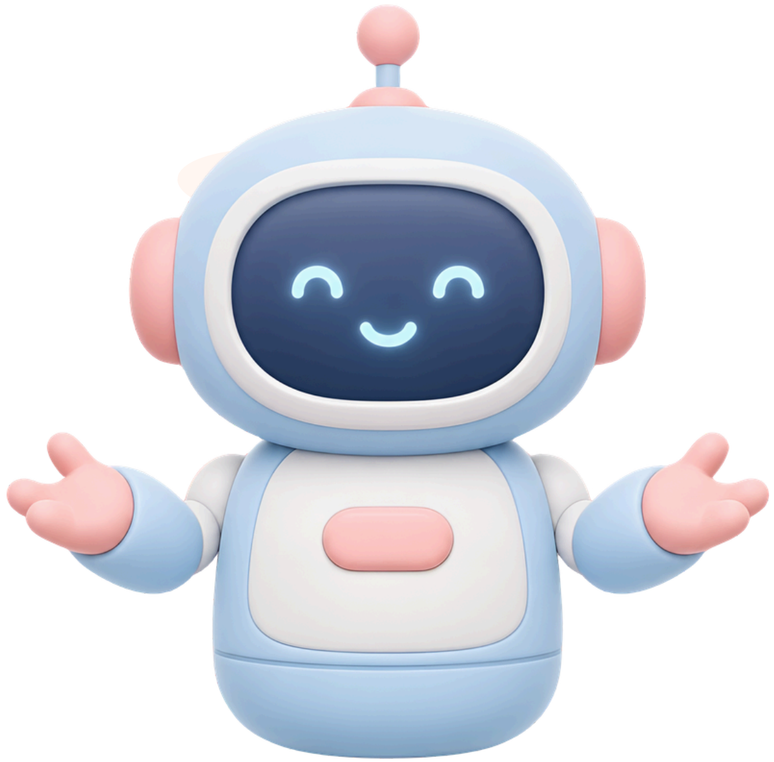
- Image Size: 128 × 128 pixels
- Split: 80% Training, 20% Validation

- Batch Size: 32Data
- Pixel Values: Normalized between 0 and 1



CNN MODEL ARCHITECTURE

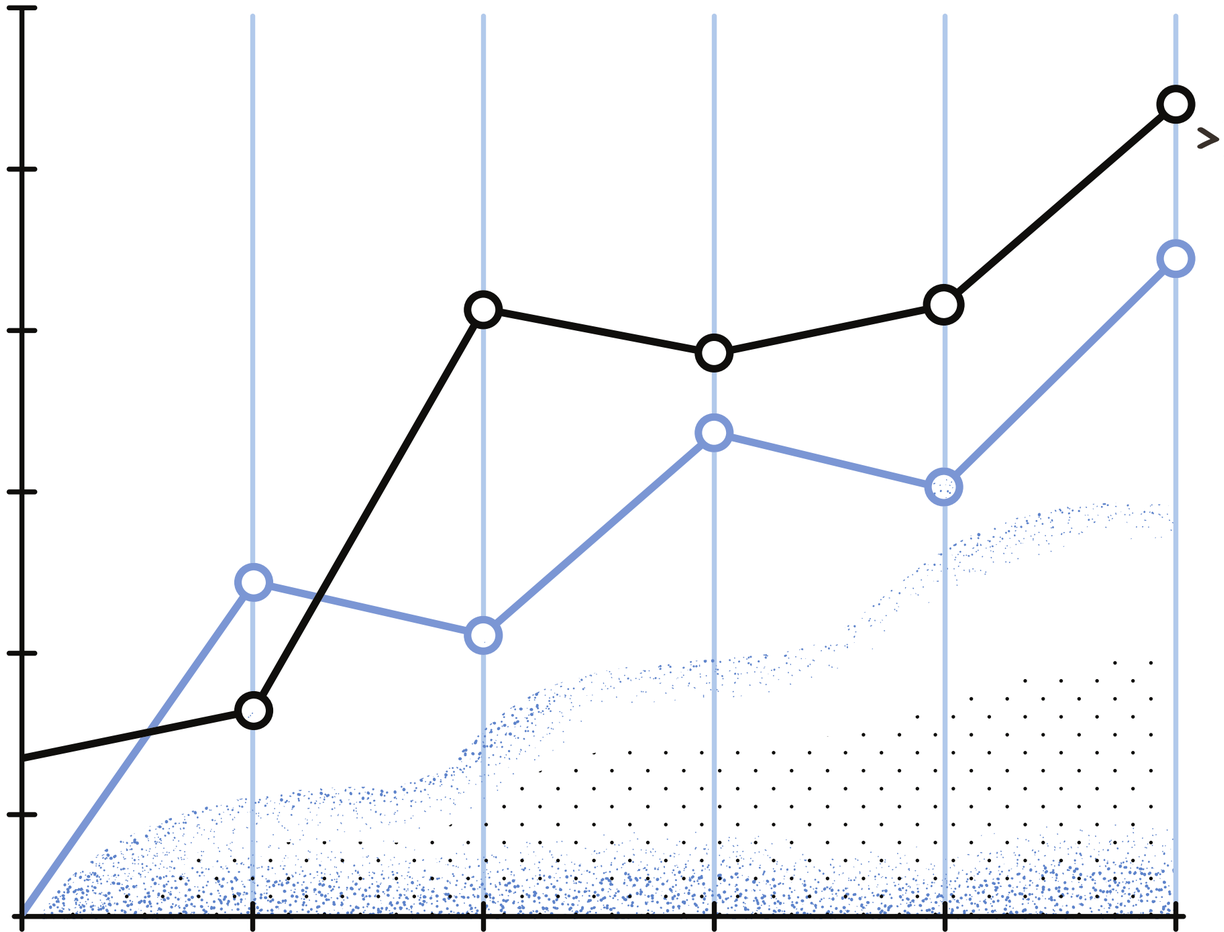
Rescaling: Normalizes image pixels.
Conv2D: Extracts features from images.
MaxPooling2D: Reduces image feature size.
Flatten: Converts features into a one-dimensional array.
Dense: Learns patterns and classifies images.
Softmax: Predicts one of 15 classes.



MODEL TRAINING

- Model Training
- Model: Convolutional Neural Network (CNN)
- Optimizer: Adam
- Loss Function: Sparse Categorical Crossentropy
- Maximum Epochs: 5 per training run
- Early Stopping: Helps prevent overfitting.

MODEL RESULTS



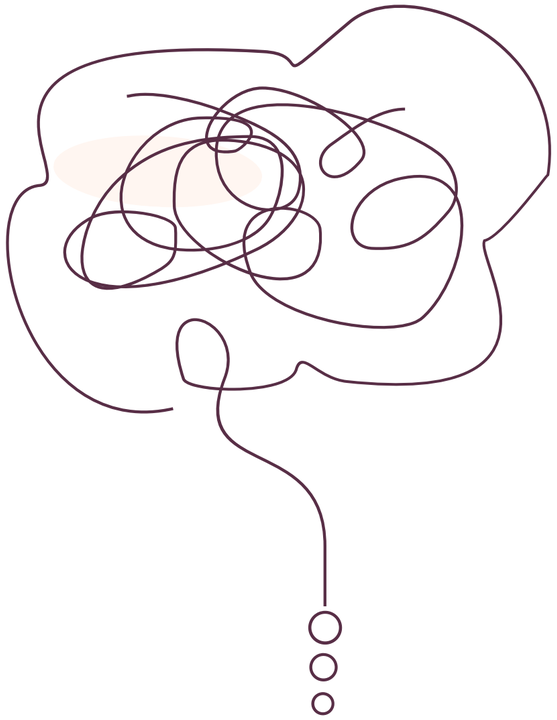
>INITIAL VALIDATION ACCURACY: 83.62%

>FINAL VALIDATION ACCURACY: 88.66%

>FINAL VALIDATION LOSS: 0.4618

PREDICTION DEMO

INPUT: PLANT LEAF IMAGE
MODEL: TRAINED CNN
PREDICTED CLASS: PEPPER BELL - BACTERIAL SPOT
RESULT: PREDICTED CLASS MATCHED THE IMAGE'S DATASET LABEL



CONCLUSION

- Built a CNN model for plant disease classification.
- Used the PlantVillage dataset with 15 classes
- Achieved 88.66% validation accuracy.
- Successfully demonstrated plant class
- prediction.

THE RESULTS



Predicted Disease:
Pepper__bell___Bacterial_spot




THANK
YOU